PRACTICAL - 1

Aim: To understand and demonstrate the use of basic commands in different operating systems (Windows, Linux, and UNIX) for managing files, directories, permissions, and user interactions through a terminal or command-line interface.

Que.1. Display the user manual of any command that we can run on the terminal.

```
C:\Users\vidhi>help
For more information on a specific command, type HELP command-name
ASSOC          Displays or modifies file extension associations.
ATTRIB         Displays or changes file attributes.
BREAK          Sets or clears extended CTRL+C checking.
BCDEDIT        Sets properties in boot database to control boot loading.
CACLS          Displays or modifies access control lists (ACLs) of files.
CALL           Calls one batch program from another.
CD             Displays the name of or changes the current directory.
CHCP           Displays or sets the active code page number.
CHDIR          Displays the name of or changes the current directory.
CHKDSK         Checks a disk and displays a status report.
CHKNTFS        Displays or modifies the checking of disk at boot time.
CLS            Clears the screen.
CMD            Starts a new instance of the Windows command interpreter.
COLOR          Sets the default console foreground and background colors.
COMP           Compares the contents of two files or sets of files.
COMPACT        Displays or alters the compression of files on NTFS partitions.
CONVERT        Converts FAT volumes to NTFS.  You cannot convert the
               current drive.
COPY           Copies one or more files to another location.
DATE           Displays or sets the date.
DEL            Deletes one or more files.
DIR            Displays a list of files and subdirectories in a directory.
DISKPART       Displays or configures Disk Partition properties.
DOSKEY         Edits command lines, recalls Windows commands, and
               creates macros.
DRIVERQUERY    Displays current device driver status and properties.
ECHO           Displays messages, or turns command echoing on or off.
ENDLOCAL       Ends localization of environment changes in a batch file.
ERASE          Deletes one or more files.
EXIT           Quits the CMD.EXE program (command interpreter).
FC             Compares two files or sets of files, and displays the
               differences between them.
FIND           Searches for a text string in a file or files.
FINDSTR        Searches for strings in files.
FOR            Runs a specified command for each file in a set of files.
FORMAT         Formats a disk for use with Windows.
FSUTIL         Displays or configures the file system properties.
FTYPE          Displays or modifies file types used in file extension
```

```
Command Prompt
FTYPE           Displays or modifies file types used in file extension
                associations.
GOTO            Directs the Windows command interpreter to a labeled line in
                a batch program.
GPRESULT        Displays Group Policy information for machine or user.
HELP            Provides Help information for Windows commands.
ICACLS          Display, modify, backup, or restore ACLs for files and
                directories.
IF              Performs conditional processing in batch programs.
LABEL           Creates, changes, or deletes the volume label of a disk.
MD              Creates a directory.
MKDIR           Creates a directory.
MKLINK          Creates Symbolic Links and Hard Links
MODE            Configures a system device.
MORE            Displays output one screen at a time.
MOVE            Moves one or more files from one directory to another
                directory.
OPENFILES       Displays files opened by remote users for a file share.
PATH            Displays or sets a search path for executable files.
PAUSE           Suspends processing of a batch file and displays a message.
POPD            Restores the previous value of the current directory saved by
                PUSHD.
PRINT           Prints a text file.
PROMPT          Changes the Windows command prompt.
PUSHD           Saves the current directory then changes it.
RD              Removes a directory.
RECOVER         Recovers readable information from a bad or defective disk.
REM             Records comments (remarks) in batch files or CONFIG.SYS.
REN             Renames a file or files.
RENAME          Renames a file or files.
REPLACE         Replaces files.
RMDIR           Removes a directory.
ROBOCOPY        Advanced utility to copy files and directory trees
SET             Displays, sets, or removes Windows environment variables.
SETLOCAL        Begins localization of environment changes in a batch file.
SC              Displays or configures services (background processes).
SCHTASKS        Schedules commands and programs to run on a computer.
SHIFT           Shifts the position of replaceable parameters in batch files.
SHUTDOWN        Allows proper local or remote shutdown of machine.
SORT            Sorts input.
START           Starts a separate window to run a specified program or command.
```

```
SHUTDOWN       Allows proper local or remote shutdown of machine.
SORT           Sorts input.
START          Starts a separate window to run a specified program or command.
SUBST          Associates a path with a drive letter.
SYSTEMINFO     Displays machine specific properties and configuration.
TASKLIST       Displays all currently running tasks including services.
TASKKILL       Kill or stop a running process or application.
TIME           Displays or sets the system time.
TITLE          Sets the window title for a CMD.EXE session.
TREE           Graphically displays the directory structure of a drive or
               path.
TYPE           Displays the contents of a text file.
VER            Displays the Windows version.
VERIFY         Tells Windows whether to verify that your files are written
               correctly to a disk.
VOL            Displays a disk volume label and serial number.
XCOPY          Copies files and directory trees.
WMIC           Displays WMI information inside interactive command shell.

For more information on tools see the command-line reference in the online help.
```

2. Change the current working directory of the user

```
C:\Users\vidhi>cd Downloads
```

3. Lists all the contents in the current working directory

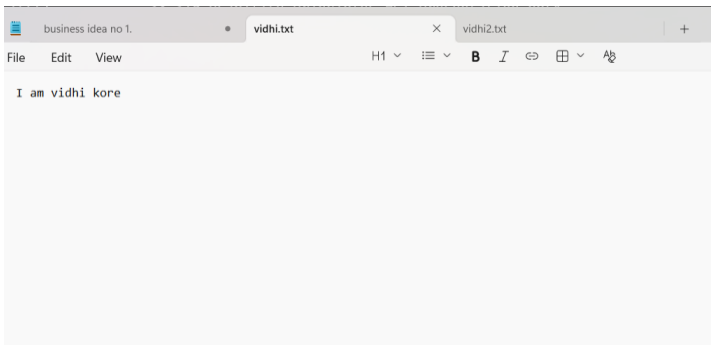
```
C:\Users\vidhi\Downloads>dir
 Volume in drive C is Windows
 Volume Serial Number is D0E0-F627

 Directory of C:\Users\vidhi\Downloads

19-01-2026  10:44    <DIR>          .
18-01-2026  19:28    <DIR>          ..
15-01-2026  13:39            60,244 Black and White Minimalist Professional Initial Logo.zip
18-09-2025  01:34           679,360 CASTE VALIDITY.pdf
13-01-2026  22:52         3,305,360 CD24061_Database%20Mana_lab_record_2026_01_13.pdf
15-01-2026  13:59           454,579 ChatGPT Image Jan 15, 2026, 01_59_55 PM.png
26-12-2025  13:16        93,695,464 CodeTantra.SEA-x64-4.2.0 (1).exe
03-01-2026  17:45        93,695,464 CodeTantra.SEA-x64-4.2.0 (2).exe
23-12-2025  19:15        93,695,464 CodeTantra.SEA-x64-4.2.0.exe
15-01-2026  22:57       100,501,080 CodeTantra.SEA-x64-4.3.0 (1).exe
17-01-2026  23:43       100,501,080 CodeTantra.SEA-x64-4.3.0 (2).exe
13-01-2026  22:50       100,501,080 CodeTantra.SEA-x64-4.3.0.exe
18-01-2026  19:22           589,064 Database-Architecture.pdf
18-01-2026  19:20         2,542,241 DBMSUNITINotes.pdf
20-08-2025  16:49               908 Desktop.lnk
30-10-2025  18:46         1,071,596 Digital.Marketing.TAE.1.Report[1].pdf
16-01-2026  18:19        66,292,520 Git-2.52.0-64-bit.exe
17-01-2026  11:49           100,194 git-cheat-sheet-education.pdf
13-01-2026  22:12           318,055 hb.pdf
14-01-2026  23:37        54,929,374 IMG_3504.MOV
30-12-2025  19:22           430,543 Journal Entries Practice Numericals.pdf
09-01-2026  17:49         1,811,413 Kaagaz_20260106_1331032235721.pdf
26-12-2025  17:51           458,394 MDM FoA Unit - 1 (1).pdf
26-12-2025  17:57           809,292 N-01_4 SEM CSE(DS) T.T.PDF
26-12-2025  17:47         2,081,547 OSIVSemUnit1.pdf
16-01-2026  23:09         4,731,520 OSLABMANUAL-DSNew.pdf
19-01-2026  10:44           189,801 palak evs.pdf
16-01-2026  23:10           464,554 PracticalIndexPageOS.pdf
17-01-2026  23:28         1,745,864 prelab output (1).pdf
17-01-2026  11:57         2,353,722 Prelab.pdf
16-01-2026  23:38           886,588 PRELABDBMS (1).pdf
13-01-2026  22:49           886,588 PRELABDBMS.pdf
09-01-2026  19:12            42,133 Purple and White Clean and Professional Resume (1) (1).pdf
09-01-2026  19:12            42,133 Purple and White Clean and Professional Resume (1) (2).pdf
09-01-2026  19:11            42,133 Purple and White Clean and Professional Resume (1).pdf
```

4. Read, modify or concatenate text files • Read a text file:

```
C:\Users\vidhi\Downloads>type vidhi.txt
 I am vidhi kore
```

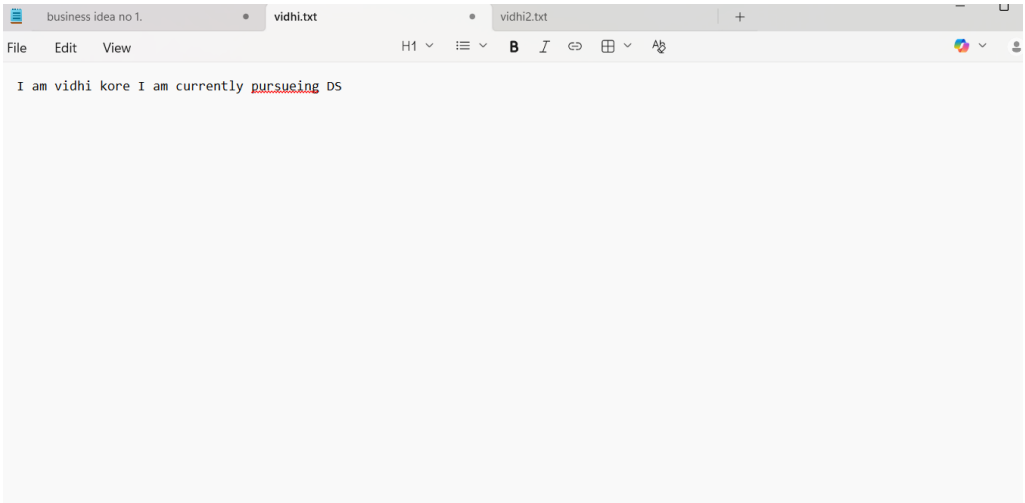
- Modify a text file:

```
C:\Users\vidhi\Downloads>notepad vidhi.txt
```



- Concatenate text files:

```
C:\Users\vidhi\Downloads>vidhi.txt vidhi2.txt > new.txt

C:\Users\vidhi\Downloads>
```



5. Create a new directory under any directory

```
C:\Users\vidhi\Downloads>mkdir ZenMode
```

6. Display the current working directory of the terminal

```
C:\Users\vidhi\Downloads>cd
C:\Users\vidhi\Downloads
```

7. Write its arguments to standard output

```
C:\Users\vidhi\Downloads>echo hello world
hello world
```

9. Delete a directory

```
C:\Users\vidhi\Downloads>del ZenMode
C:\Users\vidhi\Downloads\ZenMode\*, Are you sure (Y/N)? y
```

10. Copy a file or directory

```
C:\Users\vidhi\Downloads>copy C:\Users\vidhi\Downloads\ZenMode
C:\Users\vidhi\Downloads\ZenMode\New.txt
        1 file(s) copied.
```

11.Switch to root-user

```
C:\Users\vidhi\Downloads>runas /user:administrator cmd
Enter the password for administrator:
Attempting to start cmd as user "VIDHIKORE\administrator" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.
```

12. Moves files and directories from one directory to another

```
C:\Users\vidhi\Downloads>move C:\Users\vidhi\Downloads\ZenMode
        1 dir(s) moved.
```

13. Search for a particular string/word in a text file

```
C:\Users\vidhi\Downloads> find "Hello" New.txt

---------- NEW.TXT
```

14.Print the top N number of data of the given input

```
C:\Users\vidhi\Downloads>more New.txt
```

15. Print the last N number of data of the given input

```
C:\Users\vidhi\Downloads>more New.txt
```

16.Remove read permission from owner of the file

```
C:\Users\vidhi\Downloads>attrib -r New.txt

C:\Users\vidhi\Downloads>
```

17. Add read permission to owner plus re move write permission and add write permission to group

```
C:\Users\vidhi\Downloads>icalcs C:\Users\vidhi\Downloads
'icalcs' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\vidhi\Downloads>icacls C:\Users\vidhi\Downloads
C:\Users\vidhi\Downloads NT AUTHORITY\SYSTEM:(OI)(CI)(F)
                         BUILTIN\Administrators:(OI)(CI)(F)
                         VIDHIKORE\vidhi:(OI)(CI)(F)

Successfully processed 1 files; Failed processing 0 files
```

18. Add write permission to owner no permission to others

```
C:\Users\vidhi\Downloads>icacls ZenMode /inheritance:r /grant vidhi:W /remov
e Everyone
processed file: ZenMode
Successfully processed 1 files; Failed processing 0 files

C:\Users\vidhi\Downloads>
```

19. Assign permission to users

```
C:\Users\vidhi\Downloads>icacls ZenMode /grant users:F
processed file: ZenMode
Successfully processed 1 files; Failed processing 0 files
```

20. Assign R/W/X to others

```
C:\Users\vidhi\Downloads>icacls ZenMode /grant Everyone:(R,W,X)
processed file: ZenMode
Successfully processed 1 files; Failed processing 0 files
```

21.Remove all permission from all users

```
C:\Users\vidhi\Downloads>icacls ZenMode /inheritance:r /remove Everyone
processed file: ZenMode
Successfully processed 1 files; Failed processing 0 files
```

22. Remove read permission from user using absolute mode

```
C:\Users\vidhi\Downloads>icacls ZenMode /deny vidhi:R
processed file: ZenMode
Successfully processed 1 files; Failed processing 0 files
```

23. Add R/w permission to owner no permission to group and no permission to other using absolute mode

```
C:\Users\vidhi\Downloads>icacls ZenMode /inheritance:r /grant vidhi:RW /remo
ve Users /remove Everyone
processed file: ZenMode
Successfully processed 1 files; Failed processing 0 files
```

24. Add execute permission to owner, read permission to group and other

```
C:\Users\vidhi\Downloads>icacls ZenMode /grant %USERNAME%:(X) Users:(R) Ever
yone:(R)
processed file: ZenMode
Successfully processed 1 files; Failed processing 0 files

C:\Users\vidhi\Downloads>
```

25. Add execute permission to all user

```
C:\Users\vidhi\Downloads>icacls ZenMode /grant Everyone:(X)
processed file: ZenMode
Successfully processed 1 files; Failed processing 0 files
```